

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Scenario Based Questions

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4	Assessment:	Assessment Tool 1– Scenario Based Questions
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	B C Yugesh	Reg. No.:	192424282
Section / Batch:	2024	Date:	24/08/2026

Code of Conduct

I **B C Yugesh (192424282)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B C Yugesh

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Hypothesis Testing Using p-value – Application Response Time

Given:

- Claimed average response time = less than 2 seconds
- p-value = 0.004
- Significance level, $\alpha = 0.01$

Step 1: Hypotheses

- $H_0: \mu \geq 2$ seconds
- $H_1: \mu < 2$ seconds

Here, H_1 represents the company's claim.

Step 2: Compare p-value with α

Since,

we reject H_0 .

Step 3: Conclusion

There is sufficient statistical evidence at the 1% significance level to conclude that the average response time of the application is less than 2 seconds.

2. Hypothesis Testing Using p-value – Advertisement

Given:

- Previous click-through rate = 6%
- p-value = 0.12
- Significance level, $\alpha = 0.05$

Step 1: Hypotheses

- **H₀:** $p \leq 6\%$
- **H₁:** $p > 6\%$

The alternative hypothesis represents the claim that the new advertisement increases the click-through rate.

Step 2: Compare p-value with α

Since,

we fail to reject H₀.

Step 3: Conclusion

There is insufficient statistical evidence at the 5% significance level to conclude that the new advertisement increases the click-through rate above 6%.

3. Hypothesis Testing Using p-value – Classification Model

Given:

- Old model accuracy = 82%
- New model accuracy = 85%
- p-value = 0.01
- Significance level, $\alpha = 0.05$

Step 1: Hypotheses

- **H₀:** The new model is not significantly better than the old model.
- **H₁:** The new model is significantly better than the old model.

Step 2: Meaning of p-value

The p-value is **0.01**, which means that if there were no actual improvement in the new model, obtaining a result this extreme would be very unlikely under the statistical test.

Step 3: Compare p-value with α

Therefore, we **reject H_0** .

Step 4: Accuracy comparison

The new model has a **3 percentage-point higher accuracy**.

4. Confidence Interval + p-value – Factory Defect Rate

Given:

- Company's claimed defect rate = 2%
- 95% confidence interval = (2.4%, 3.6%)
- p-value = 0.01
- Significance level, $\alpha = 0.05$

Step 1: Hypotheses

- **H_0 :** $p = 2\%$
- **H_1 :** $p \neq 2\%$

Step 2: Interpret the confidence interval

The 95% confidence interval is:

The claimed value of 2% is not inside the confidence interval.

Also, the entire interval is above 2%.

Therefore, the actual defect rate is likely to be higher than 2%.

Step 3: Use the p-value

Since,

we reject H_0 .

Conclusion

Both the confidence interval and p-value provide evidence against the company's claim.

Therefore, the company's claim that the defect rate is 2% should be rejected. The actual defect rate is likely to be higher than 2%, with a plausible range of 2.4% to 3.6%.

5. Confidence Interval + p-value – New UI Design

Given:

- 95% confidence interval = (₹120, ₹480)
- p-value = 0.002
- Significance level, $\alpha = 0.05$

Step 1: Hypotheses

- **H_0 :** The new UI design does not increase average customer spending.
- **H_1 :** The new UI design increases average customer spending.

Step 2: Interpret the confidence interval

The 95% confidence interval is:

This indicates that the new UI design is estimated to increase average spending by between ₹120 and ₹480 per customer.

Since the entire interval is **positive** and does not contain ₹0, there is evidence of an increase in spending.

Step 3: Use the p-value

Since,

we reject H_0 .

Step 4: Business Recommendation

The statistical results indicate that the new UI design significantly increases average customer spending.

The expected increase is between ₹120 and ₹480 per customer.

Therefore, the company should consider implementing the new UI design, provided the expected additional revenue is greater than the implementation and maintenance cost.